

11.4 Build or Bust? (BLD)

A study of some aspects of building design, including withstanding earthquake damage, vibration isolation and sound-proofing:

- earthquake detection
- vibration and resonance in structures
- damping vibration using ductile materials.

The behaviour of oscillators is modelled using the mathematics of simple harmonic motion, and physical models are used to explore the behaviour of structures.

There are opportunities to develop ICT skills using data logging and spreadsheets.

Students will be assessed on their ability to:	Suggested experiments
119 recall that the condition for simple harmonic motion is $F = -kx$, and hence identify situations in which simple harmonic motion will occur	
120 recognise and use the expressions $a = -\omega^2 x$, $a = -A\omega^2 \cos \omega t$, $v = A\omega \sin \omega t$, $x = A \cos \omega t$ and $T = 1/f = 2\pi/\omega$ as applied to a simple harmonic oscillator	
121 obtain a displacement – time graph for an oscillating object and recognise that the gradient at a point gives the velocity at that point	Use a motion sensor to generate graphs of SHM
122 recall that the total energy of an undamped simple harmonic system remains constant and recognise and use expressions for total energy of an oscillator	
123 distinguish between free, damped and forced oscillations	

Students will be assessed on their ability to:	Suggested experiments
124 investigate and recall how the amplitude of a forced oscillation changes at and around the natural frequency of a system and describe, qualitatively, how damping affects resonance	Use, for example, a vibration generator to investigate forced oscillations
125 explain how damping and the plastic deformation of ductile materials reduce the amplitude of oscillation	Use, for example, a vibration generator to investigate damped oscillations